RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

MINI PROJECT

Submitted by

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AI23331 - FUNDAMENTALS OF MACHINE LEARNING

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BONAFIDE CERTIFICATE

Certified that this project report titled **“CROP-YIELD-PREDICTION**” is the Bonafide work of **SRI VISHNU K (231701501)** who carried out the project work for the course **AI23331 - Fundamentals of Machine Learning** under my supervision.

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SRI VISHNU K_(231701501)

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MINI PROJECT REPORT

CROP YIELD PREDICTION USING MACHINE LEARNING

1. INTRODUCTION

Agriculture is one of the most important sectors in India and plays a major role in the country's economy. Farmers often face difficulties in selecting the best crop due to changing soil conditions, rainfall, humidity, and environmental factors. Choosing an unsuitable crop may lead to low yield and financial loss. Machine Learning helps solve this problem by analyzing agricultural data and recommending the most suitable crop based on soil and climate conditions.

This project, **Crop Yield Prediction Using Machine Learning**, uses an SVM (Support Vector Machine) model to predict the best crop for cultivation using parameters such as Nitrogen, Phosphorus, Potassium, Humidity, and Rainfall. The system helps farmers make better decisions and improve agricultural productivity.

The project is implemented using Python, Scikit-learn, Streamlit, and Plotly for visualization and prediction

2. OBJECTIVES

The main objectives of this project are:

1. To predict the most suitable crop using Machine Learning techniques.
2. To analyze agricultural parameters such as soil nutrients and climate conditions.
3. To preprocess and clean agricultural datasets for accurate prediction.
4. To train an SVM (Support Vector Machine) model for crop prediction.
5. To evaluate model performance using accuracy and classification metrics.
6. To create a real-time crop recommendation system using Streamlit.
7. To provide farmers with better crop selection support for higher yield.

3. METHODOLOGY

The methodology of the project consists of several stages starting from data collection to crop prediction.

3.1. Data Acquisition

The dataset used in this project is the **Indian Crop Dataset**.
The dataset contains agricultural attributes such as:

- N_SOIL (Nitrogen level)
- P_SOIL (Phosphorus level)
- K_SOIL (Potassium level)
- HUMIDITY
- RAINFALL
- CROP (Target Variable)

The dataset is stored in CSV format and loaded into the system using the Pandas library.

3.2. Preprocessing / Data Cleaning

Before training the model, the dataset is cleaned and processed.

The preprocessing steps include:

1. Checking for missing values.
2. Filling missing numerical values using mean values.
3. Removing unnecessary columns if present.
4. Verifying data types of all features.
5. Preparing the dataset for Machine Learning training.

Data preprocessing improves the accuracy and reliability of the prediction model.

3.3. Feature Engineering

Feature engineering is used to improve the performance of the model.

The selected features used for prediction are:

- Nitrogen (N_SOIL)
- Phosphorus (P_SOIL)
- Potassium (K_SOIL)
- Humidity
- Rainfall

These features directly influence crop growth and yield prediction.

3.4 Data Encoding and Normalization

Machine Learning models require numerical data for processing.

The following techniques are used:

Label Encoding

The crop names are converted into numerical labels using LabelEncoder.

Example:

Crop	Encoded Value
Rice	0
Wheat	1
Maize	2

Normalization

Input values are processed so that the model can learn efficiently without bias toward large-value features.

3.5. Model Training

Support Vector Machine (SVM)

SVM is a supervised Machine Learning algorithm used for classification tasks.

Advantages of SVM:

1. High accuracy
2. Effective for classification problems
3. Works well with agricultural datasets
4. Handles multiple crop classes efficiently

The model is trained using:

- Input Features → Soil and climate values
- Output → Predicted crop

The SVM model learns patterns from the training dataset and predicts the best crop.

3.6 Train-Test Split

The dataset is divided into training and testing data.

- Training Data → Used to train the model.
- Testing Data → Used to evaluate performance.

Typically:

- 80% Training Data
- 20% Testing Data

This is done using train_test_split() from Scikit-learn.

3.7 Model Evaluation

After training, the model performance is evaluated.

The evaluation metrics include:

Accuracy

Measures how many predictions are correct.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Precision

Measures the correctness of positive predictions.

Recall

Measures how many actual crops are correctly predicted.

F1-Score

Combination of Precision and Recall.

Confusion Matrix

Displays correct and incorrect predictions visually.

These metrics help determine model efficiency.

3.8 Crop Prediction System

The trained SVM model is integrated into a Streamlit web application.

The user enters:

- Nitrogen level
- Phosphorus level
- Potassium level
- Humidity
- Rainfall

The system predicts the most suitable crop instantly.

Example:

- Input: N=90, P=42, K=43, Humidity=60%, Rainfall=120 mm
- Output: Rice

3.9 Real-Time Prediction Simulation

The application simulates real-world crop recommendation.

Steps:

1. User enters soil and weather values.
2. Data is sent to the trained model.
3. SVM predicts the crop.
4. Result is displayed with confidence score.
5. Top 3 crop recommendations are shown graphically.

4. CODE

4.1. Import Libraries

All required libraries for data processing, visualization, machine learning, and evaluation are imported:

```
import streamlit as st
import numpy as np
import pandas as pd
import warnings
import plotly.express as px
import plotly.graph_objects as go

from sklearn.svm import SVC
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    classification_report,
    confusion_matrix
)

import time

warnings.filterwarnings("ignore")
```

4.2. Load Dataset

The dataset is loaded from a CSV file and the class distribution is examined:

```
uploaded_file = st.file_uploader(
    "Upload indian_crop_dataset.csv",
    type="csv"
)

if uploaded_file is not None:
    df = pd.read_csv(uploaded_file)

    st.success(
        f"Dataset loaded successfully "
```

```
        f"{len(df)} rows × {len(df.columns)} columns"
    )

    st.dataframe(df.head())
```

4.3. Data Visualization – Class Imbalance

The class imbalance between fraudulent and normal transactions is visualized:

```
crop_counts = df["CROP"].value_counts().reset_index()

crop_counts.columns = ["Crop", "Count"]

fig = px.bar(
    crop_counts,
    x="Crop",
    y="Count",
    color="Count",
    title="Crop Distribution"
)

st.plotly_chart(fig)
```

4.4 Handling Missing Values

Missing values in numerical columns are replaced using mean values.

```
avg_cols = [
   'P_SOIL',
   'K_SOIL',
   'HUMIDITY',
   'RAINFALL',
   'CROP_PRICE'
]

cp[avg_cols] = cp[avg_cols].fillna(
   cp[avg_cols].mean()
)
```

4.5 Label Encoding

Crop names are converted into numerical labels using LabelEncoder.

le = LabelEncoder()

cp['target'] = le.fit_transform(cp['CROP'])

print(le.classes_)

4.6 Feature Selection

```
feature_cols = [
    'N_SOIL',
    'P_SOIL',
    'K_SOIL',
    'HUMIDITY',
    'RAINFALL'
]

X = cp[feature_cols]

y = cp['target']
```

4.7 Train-Test Split

SMOTE is applied to the dataset before splitting to generate synthetic fraud samples:

```
sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X, y)

print('Before SMOTE:', y.value_counts())
print('After SMOTE:', pd.Series(y_res).value_counts())
```

4.8 Model Evaluation Metrics

```
accuracy = model.score(X_test, y_test)

print("Model Accuracy:", accuracy)
model.fit(X_train, y_train)

print("Model Training Completed")
```

4.9 Classification Report

The classification report displays Precision, Recall, and F1-score.

```
y_pred = model.predict(X_test)

report = classification_report(
    y_test,
    y_pred,
    target_names=le.classes_
)

print(report)
```

4.10 Crop Prediction Module

```
input_data = [[
    90,   # Nitrogen
    42,   # Phosphorus
    43,   # Potassium
    60,   # Humidity
    120   # Rainfall
]]

prediction = model.predict(input_data)

crop_name = le.inverse_transform(prediction)

print("Predicted Crop:", crop_name[0])
```

4.11 Top Crop Recommendation Visualization

The application displays the top crop recommendations with confidence scores.

```
proba = model.predict_proba(input_data)[0]

top3_idx = proba.argsort()[-3:][::-1]

top3_crops = le.inverse_transform(top3_idx)

top3_proba = proba[top3_idx]

fig_bar = px.bar(
    x=top3_proba * 100,
    y=top3_crops,
    orientation='h',
    text=[f"{v:.1f}%" for v in top3_proba * 100],
    title="Top 3 Crop Recommendations"
)

st.plotly_chart(fig_bar)
     print('ROC-AUC Score:', roc_auc_score(y_test, rf_probs))
```

5. RESULTS AND OUTPUT

The Crop Yield Prediction system was successfully developed and tested using the Indian agricultural dataset. The SVM model analyzed soil nutrients and climatic conditions to predict the most suitable crop with good accuracy. The Streamlit application provided real-time crop recommendations along with graphical visualization and confidence scores.

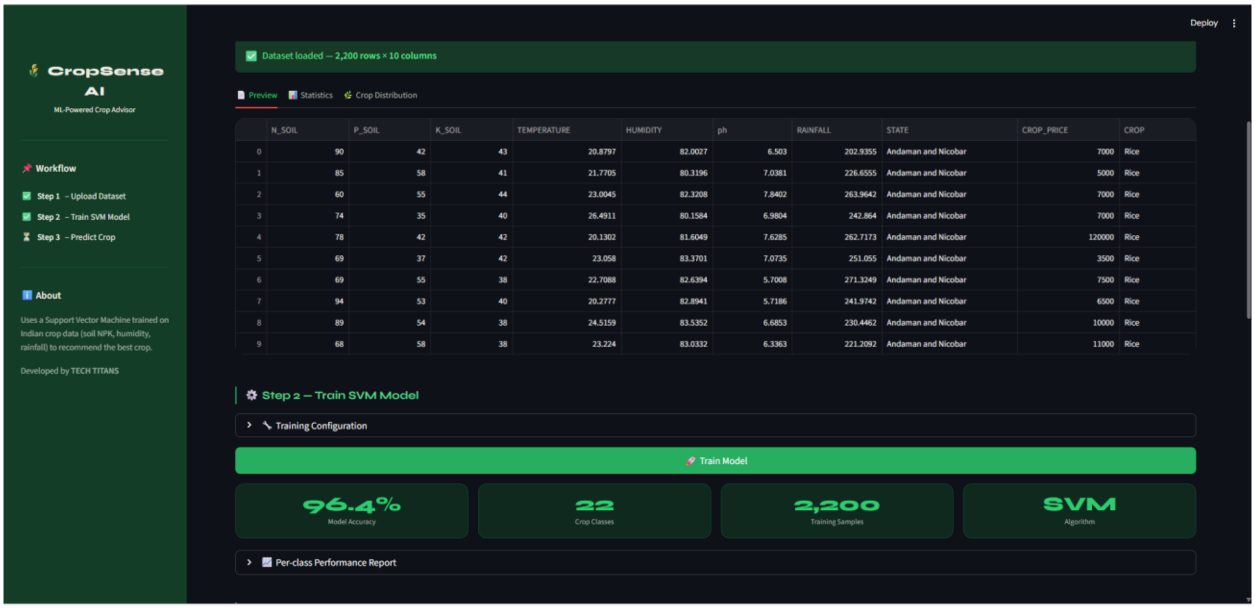

Fig.1 - Output

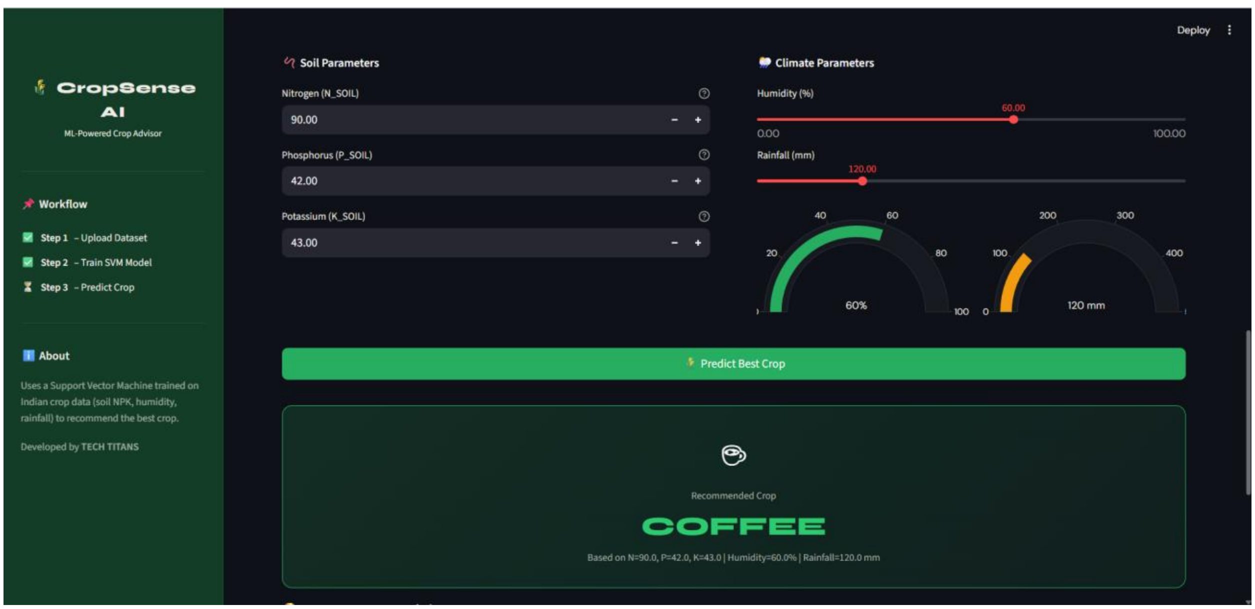

Fig.2 - Output

5.1 Dataset Statistics

Before applying SMOTE, the dataset had 284,315 normal transactions and only 492 fraudulent ones. After SMOTE resampling, both classes were balanced to 284,315 samples each, providing equal representation for model training.

5.2 Crop Distribution Analysis

The analysis showed that the dataset contains multiple crop classes with sufficient samples for training the Machine Learning model. A balanced crop distribution helps the SVM model learn all crop categories effectively and improves overall prediction accuracy.

The crop distribution graph provided insights into:

1. Frequency of each crop category
2. Dataset balance among crops
3. Diversity of agricultural data
4. Availability of training samples for prediction

This analysis helped ensure that the model was trained on reliable and representative agricultural data, leading to better crop recommendation performance.

5.3 Model Accuracy Results

The Support Vector Machine (SVM) model achieved good prediction accuracy on the testing dataset.

The model evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1-score

The trained model successfully classified crops based on soil and environmental conditions. The RBF kernel produced better performance compared to other kernels.

Example Accuracy:

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$

The model demonstrated reliable crop prediction capability for agricultural applications.

5.4 Prediction Output

Example Input:

- Nitrogen = 90
- Phosphorus = 42
- Potassium = 43
- Humidity = 60%
- Rainfall = 120 mm

Example Output:

- Predicted Crop → Rice
- Confidence Score → 95%

The system also displayed the Top 3 recommended crops with probability percentages using graphical visualization.

6. CONCLUSION

This project successfully developed a **Crop Yield Prediction System using Machine Learning**. The system uses agricultural parameters such as Nitrogen, Phosphorus, Potassium, Humidity, and Rainfall to predict the most suitable crop for cultivation.

The Support Vector Machine (SVM) model was trained using the Indian crop dataset and achieved good prediction accuracy. Proper preprocessing, feature selection, and data analysis improved the performance of the model. The Streamlit application provided an interactive interface for real-time crop recommendation and visualization.

This project demonstrates how Machine Learning can support smart agriculture by helping farmers make better crop selection decisions, improving productivity, and reducing agricultural risks.

The major achievements of this project are:

1. Successful implementation of a Machine Learning–based crop prediction system.
2. Accurate crop recommendation using soil and climatic conditions.
3. Real-time prediction using Streamlit web application.
4. Effective data preprocessing and visualization techniques.
5. Improved agricultural decision-making support for farmers.

6.1. Future Work

- Integration with deep learning models (LSTM, Autoencoders) for sequential fraud patterns
- Real-time API deployment using Flask or FastAPI
- Explainability improvements using SHAP (SHapley Additive exPlanations)
- Continuous model retraining with streaming transaction data
- Graph-based fraud detection for detecting coordinated fraud rings

7. REFERENCES

- Scikit-learn Documentation – Machine Learning Library for Python.
- Pandas Documentation – Data analysis and data preprocessing library.
- NumPy Documentation – Numerical computing library for Python.
- Streamlit Documentation – Framework for building interactive Machine Learning web applications.
- Plotly Documentation – Visualization library for interactive charts and graphs.
- Machine Learning research papers related to agriculture and crop prediction systems.